tolerance

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This file is part of CasADi.

CasADi -- A symbolic framework for dynamic optimization.
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1 Integrator tolerances

[1]:
```
import casadi as ca
from numpy import *
from pylab import *
```

[2]:
```
x=ca.SX.sym('x')
dx=ca.SX.sym('dx')
states = ca.vertcat(x,dx)
```

[3]:
```
dae={'x':states, 'ode':ca.vertcat(dx,-x)}
```

[4]:
```
tend = 2*pi*3
ts = linspace(0,tend,1000)
```

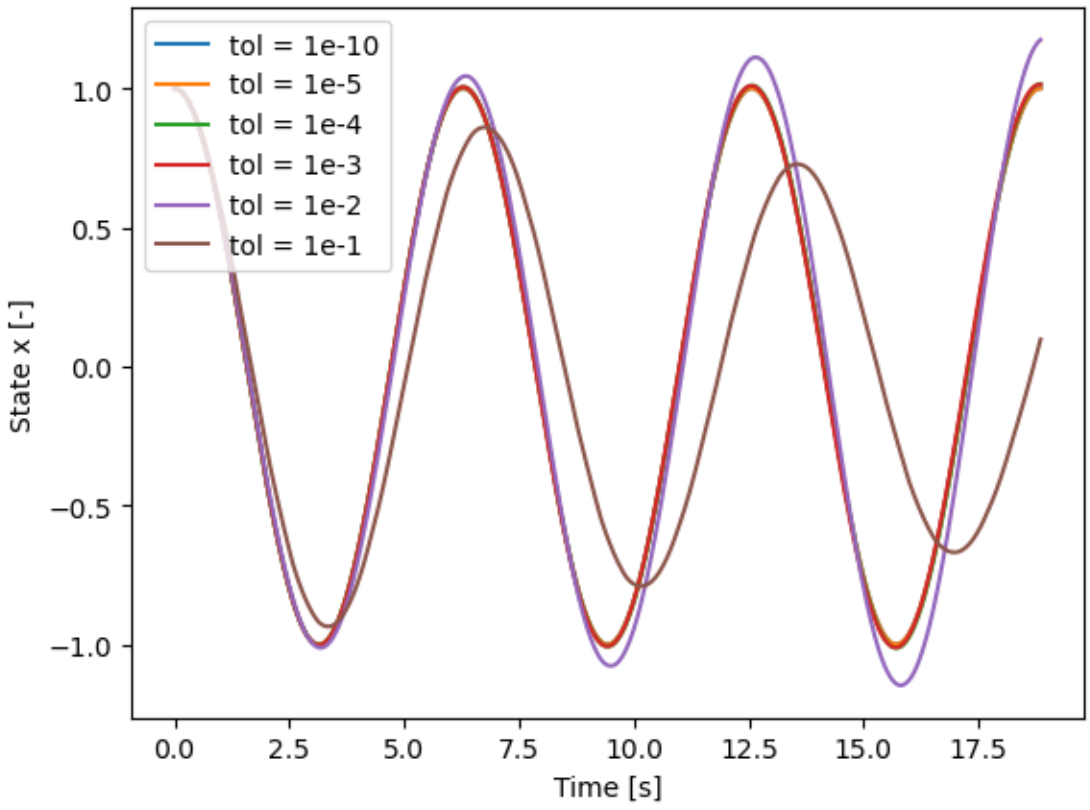
[5]:
```
tolerances = [-10,-5,-4,-3,-2,-1]
```

[6]:
```
figure()
```

[6]:
```
<Figure size 640x480 with 0 Axes>
```

```
<Figure size 640x480 with 0 Axes>
```

[7]:
```
for tol in tolerances:
  opts = {'reltol':10.0**tol, 'abstol':10.0**tol}
  F = ca.integrator('F', 'cvodes', dae, 0, ts, opts)
  res = F(x0=[1,0])
  plot(ts,array(res['xf'])[0,:].T,label='tol = 1e%d' % tol)
legend( loc='upper left')
xlabel('Time [s]')
ylabel('State x [-]')
show()
```



[8]:
```
tolerances = logspace(-15,1,500)
endresult=[]
```

[9]:
```
for tol in tolerances:
  opts = {}
```

```
  opts['reltol'] = tol
  opts['abstol'] = tol
  F = ca.integrator('F', 'cvodes', dae, 0, tend, opts)
  res = F(x0=[1,0])
  endresult.append(res['xf'][0])
```

[10]:
```
endresult = ca.vcat(endresult)
```

[11]:
```
figure()
loglog(tolerances,(array(endresult)-1),'b',label='Positive error')
loglog(tolerances,-(array(endresult)-1),'r',label='Negative error')
xlabel('Integrator relative tolerance')
ylabel('Error at the end of integration time')
legend(loc='upper left')
show()
```

